Chapter 8. Loglinear Models for Contingency Tables

Two-, three- and higher-way tables

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Loglinear models are Poisson GLMs with log link for the cell counts of a contingency table. They treat all cross-classified variables symmetrically, as joint responses, and their purpose is to describe association and interaction patterns. With one response it is simpler to use logit models; Section 8.5 shows that logit models with categorical predictors are special loglinear models. Throughout, $\{n_{ij}\}$ or $\{n_{ijk}\}$ are observed counts, $\{\mu\}$ expected counts and $\{\pi\}$ cell probabilities; a "+" subscript denotes summation over that index.

8.1 Loglinear models for two-way tables

Motivation. For an $I \times J$ table the familiar independence structure $\pi_{ij} = \pi_{i+}\pi_{+j}$ is multiplicative, so on the log scale it is additive, like a two-way ANOVA without interaction. Adding the "interaction" terms gives a model that can describe any association.

Let $\{\mu_{ij}\}$ be expected counts: either $\mu_{ij} = n\pi_{ij}$ under multinomial sampling, or $\mu_{ij} = E(Y_{ij})$ for $N = IJ$ independent Poisson counts. Formulas in terms of μ_{ij} cover both cases. Denote the row variable by X and the column variable by Y.

8.1.1 Independence model

Definition (Loglinear model of independence)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \tag{8.1}$$

with identifiability constraints such as $\lambda_I^X = \lambda_J^Y = 0$ (baseline coding) or $\sum_i \lambda_i^X = \sum_j \lambda_j^Y = 0$.

Theorem 8.1 (Independence as a loglinear model)

For positive $\{\mu_{ij}\}$ with $\mu_{ij} = n\pi_{ij}$, the variables X and Y are independent if and only if (8.1) holds for some parameters. The ML fitted values are $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$, then $\log \mu_{ij} = \log n + \log \pi_{i+} + \log \pi_{+j}$. Put $\lambda_i^X = \log \pi_{i+} - \log \pi_{I+}$, $\lambda_j^Y = \log \pi_{+j} - \log \pi_{+J}$ and $\lambda = \log n + \log \pi_{I+} + \log \pi_{+J}$; this is (8.1) with baseline constraints.

Step 2. Conversely, if $\mu_{ij} = a_i b_j$ with $a_i = e^{\lambda + \lambda_i^X}$, $b_j = e^{\lambda_j^Y}$, then $\mu_{i+} = a_i b_+$, $\mu_{+j} = a_+ b_j$, $\mu_{++} = a_+ b_+$, so

$$\pi_{ij} = \frac{a_i b_j}{a_+ b_+} = \frac{a_i b_+}{a_+ b_+} \cdot \frac{a_+ b_j}{a_+ b_+} = \pi_{i+}\pi_{+j}.$$

Step 3. The fitted values $n_{i+}n_{+j}/n$ satisfy the model and reproduce the row and column totals; by Birch's result (Theorem 8.19) they are the ML estimates. ∎

Hence the Pearson X^2 and G^2 tests of independence are goodness-of-fit tests of the loglinear model (8.1), with $\mathrm{df} = (I-1)(J-1)$.

8.1.2 Interpretation of parameters

A loglinear GLM takes the N cell counts, not the n subjects, as its observations, and does not distinguish response from explanatory variables. For interpretation it nevertheless helps to form a logit.

Proposition 8.2 (Independence means a constant logit)

Let $J = 2$. Under (8.1),

$$\operatorname{logit} P(Y = 1 \mid X = i) = \lambda_1^Y - \lambda_2^Y \qquad \text{for every } i,$$

so the odds of column 1 equal $\exp(\lambda_1^Y - \lambda_2^Y)$ in every row.

Proof. $P(Y = 1 \mid X = i)/P(Y = 2 \mid X = i) = \mu_{i1}/\mu_{i2}$, hence the logit equals

$$\log \mu_{i1} - \log \mu_{i2} = (\lambda + \lambda_i^X + \lambda_1^Y) - (\lambda + \lambda_i^X + \lambda_2^Y) = \lambda_1^Y - \lambda_2^Y,$$

which does not depend on i. ■

In general, differences of main-effect parameters for one variable are log odds of one category versus another on that variable.

8.1.3 Saturated model

Definition (Saturated two-way model)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}. \tag{8.2}$$

The association terms λ_{ij}^{XY} (coefficients of products of dummy variables) measure departures from independence; with baseline constraints $\lambda_{Ij}^{XY} = \lambda_{iJ}^{XY} = 0$.

Theorem 8.3 (Saturated model and odds ratios)

(a) Model (8.2) has IJ nonredundant parameters and describes every positive table $\{\mu_{ij}\}$ exactly.
(b) Every odds ratio is a contrast of the association parameters; for rows a, c and columns b, d,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}.$$

In particular, for a 2×2 table $\log \theta = \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY}$.
(c) Independence holds iff all $\lambda_{ij}^{XY} = 0$; the test has df $= (I - 1)(J - 1)$.

Proof.

Step 1. Counting under baseline constraints: $1 + (I - 1) + (J - 1) + (I - 1)(J - 1) = IJ$.

Step 2. Given positive $\{\mu_{ij}\}$, set $h_{ij} = \log \mu_{ij}$ and define

$$\lambda = h_{IJ}, \quad \lambda_i^X = h_{iJ} - h_{IJ}, \quad \lambda_j^Y = h_{Ij} - h_{IJ}, \quad \lambda_{ij}^{XY} = h_{ij} - h_{iJ} - h_{Ij} + h_{IJ}.$$

These satisfy the baseline constraints and $\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY} = h_{ij}$, so every positive table is represented (uniquely, since the map is invertible).

Step 3. For (b), substitute (8.2): in $h_{ab} + h_{cd} - h_{ad} - h_{cb}$ the terms λ, $\lambda_a^X, \lambda_c^X, \lambda_b^Y, \lambda_d^Y$ each appear once with $+$ and once with $-$, leaving the stated contrast.

Step 4. For (c), by step 2 all $\lambda_{ij}^{XY} = 0$ iff $h_{ij} - h_{iJ} - h_{Ij} + h_{IJ} = 0$ for all i, j, i.e. all odds ratios with the last row and column are 1, which is equivalent to independence. The $(I - 1)(J - 1)$ free association parameters give the df. ■

Hierarchical models

A model is *hierarchical* if whenever it contains a term it also contains all lower-order terms built from the same variables (e.g. λ_{ij}^{XY} requires λ_i^X, λ_j^Y). Nonhierarchical models such as $\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_{ij}^{XY}$ have fits and interpretations that depend on the coding (with baseline coding it forces $\log \mu_{Ij} = \lambda$ in every column) and are rarely sensible. In a model with λ^{XY}, interpretation should focus on the highest-order terms, not on main effects.

8.1.4 Alternative parameter constraints

Instead of $\lambda_{Ij}^{XY} = \lambda_{iJ}^{XY} = 0$ one may use $\sum_i \lambda_{ij}^{XY} = \sum_j \lambda_{ij}^{XY} = 0$. Only contrasts such as the one in Theorem 8.3(b) are invariant. For a 2×2 table with $\log \theta = 2.0$: baseline coding gives $\lambda_{11}^{XY} = 2.0$ and the others 0; sum-to-zero coding gives $\lambda_{11}^{XY} = \lambda_{22}^{XY} = 0.5$, $\lambda_{12}^{XY} = \lambda_{21}^{XY} = -0.5$. Both give $\log \theta = 2.0$. Baseline coding has the advantage that some parameters in a set may have infinite estimates while others stay finite.

8.1.5 Multinomial models for cell probabilities

Proposition 8.4 (Poisson to multinomial)

If Y_{ij} are independent Poisson with $\log \mu_{ij}$ given by (8.2), then conditional on $\sum Y_{ij} = n$ the counts are multinomial$(n, \{\pi_{ij}\})$ with

$$\pi_{ij} = \frac{\exp(\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY})}{\sum_a \sum_b \exp(\lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}^{XY})}, \tag{8.3}$$

in which λ cancels.

Proof. Let $\mu_{++} = \sum \mu_{ij}$; the total is Poisson(μ_{++}). For counts $\{n_{ij}\}$ summing to n,

$$P(\{Y_{ij} = n_{ij}\} \mid Y_{++} = n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}} / n_{ij}!}{e^{-\mu_{++}} \mu_{++}^n / n!} = \frac{n!}{\prod n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu_{++}} \Big)^{n_{ij}},$$

since $\prod e^{-\mu_{ij}} = e^{-\mu_{++}}$ and $\sum n_{ij} = n$. This is multinomial with $\pi_{ij} = \mu_{ij}/\mu_{++}$, which is (8.3); the common factor e^{λ} cancels. ∎

The intercept λ only describes the total sample size, which is random under Poisson sampling but fixed under multinomial sampling.

8.2 Loglinear models for independence and interaction in three-way tables

Motivation. Three-way tables admit several distinct kinds of independence (mutual, joint, conditional) and of association (homogeneous or not). Each one corresponds to a loglinear model in which certain interaction terms are absent, so the model symbol tells you the independence structure.

The following lemma turns every independence statement into a statement about which terms appear.

Lemma 8.5 (Generating classes and factorization)

Let a hierarchical loglinear model have generating class $\{C_1, \dots, C_m\}$ (the highest-order terms, i.e. the model symbol; e.g. (XZ, YZ) has $C_1 = \{X, Z\}$, $C_2 = \{Y, Z\}$). Then a positive table $\{\mu\}$ satisfies the model if and only if

$$\mu(\mathbf{x}) = \prod_{r=1}^{m} \psi_r(\mathbf{x}_{C_r})$$

for some positive functions ψ_r that depend on the cell $\mathbf{x}$ only through the coordinates in C_r.

Proof. [Supplement]

Step 1. Every term of the model involves a subset of some C_r, hence is a function of $\mathbf{x}_{C_r}$. Grouping terms by a generator containing them, $\log \mu = \sum_r g_r(\mathbf{x}_{C_r})$; exponentiate to get the product form.

Step 2. Conversely, by Theorem 8.3 (step 2, applied to the variables in C_r, and its obvious extension to more than two variables) any function $g(\mathbf{x}_C)$ decomposes as a constant plus main effects plus interactions of every subset of C. Since the model is hierarchical, all these subsets are terms of the model. Summing the decompositions of $\log \psi_r$ gives $\log \mu$ in the model's form. ■

8.2.1 Types of independence

Consider an $I \times J \times K$ table of three responses X, Y, Z with probabilities $\{\pi_{ijk}\}$ (or Poisson means $\{\mu_{ijk}\}$).

Definition (Three kinds of independence)

(i) *Mutual independence*: $\pi_{ijk} = \pi_{i++}\pi_{+j+}\pi_{++k}$ for all i, j, k.
(ii) *Joint independence* of Y from (X, Z): $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$, i.e. two-way independence between Y and the IK combinations of (X, Z).
(iii) *Conditional independence* of X and Y given Z: $\pi_{ij|k} = \pi_{i+|k}\pi_{+j|k}$ for all k, where $\pi_{ij|k} = P(X = i, Y = j \mid Z = k)$; equivalently

$$\pi_{ijk} = \pi_{i+k}\pi_{+jk}/\pi_{++k} \qquad \text{for all } i, j, k. \tag{8.4}$$

Theorem 8.6 (Independence structures as loglinear models)

For positive tables:
(a) mutual independence $\iff$ (X, Y, Z): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z$;
(b) Y jointly independent of (X, Z) $\iff$ (XZ, Y): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ}$;
(c) $X \perp Y \mid Z$ $\iff$ (XZ, YZ): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}$.
Moreover (a) $\Rightarrow$ (b) $\Rightarrow$ (c).

Proof. By Lemma 8.5 it suffices to match each independence statement with a product form. Write $\mu = n\pi$.

Step 1 (c). If (8.4) holds, $\mu_{ijk} = \psi_1(i,k)\psi_2(j,k)$ with $\psi_1 = n\pi_{i+k}$ and $\psi_2 = \pi_{+jk}/\pi_{++k}$. Conversely, if $\mu_{ijk} = A_{ik}B_{jk}$, then $\mu_{i+k} = A_{ik}B_{+k}$, $\mu_{+jk} = A_{+k}B_{jk}$ and $\mu_{++k} = A_{+k}B_{+k}$, whence

$$\frac{\mu_{i+k}\mu_{+jk}}{\mu_{++k}} = \frac{A_{ik}B_{+k}\, A_{+k}B_{jk}}{A_{+k}B_{+k}} = A_{ik}B_{jk} = \mu_{ijk},$$

which is (8.4) after dividing by n.

Step 2 (b). If $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$ we have a product $\psi_1(i,k)\psi_2(j)$. Conversely if $\mu_{ijk} = A_{ik}B_j$ then $\mu_{i+k} = A_{ik}B_+$, $\mu_{+j+} = A_{++}B_j$, $\mu_{+++} = A_{++}B_+$, and $\mu_{i+k}\mu_{+j+}/\mu_{+++} = A_{ik}B_j$, i.e. $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$.

Step 3 (a). Same computation with $\mu_{ijk} = A_iB_jC_k$: $\mu_{i++}\mu_{+j+}\mu_{++k}/\mu_{+++}^2 = A_iB_jC_k$.

Step 4. The implications follow because the model in (a) is a special case of (b) (set $\lambda^{XZ} = 0$), which is a special case of (c) (set $\lambda^{YZ} = 0$). ■

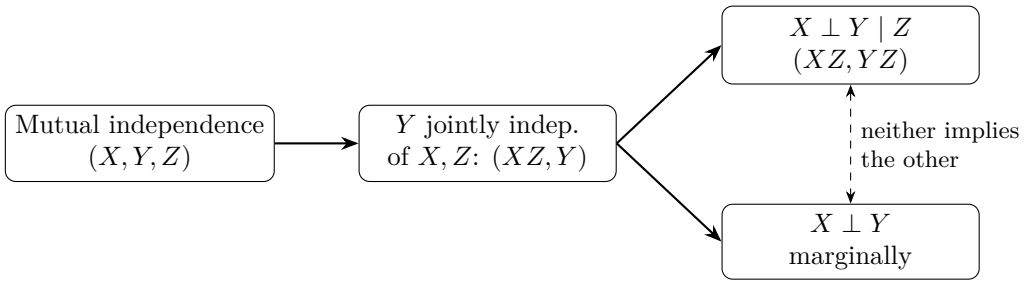


Figure 1: Relationships among types of XY independence (book Figure 8.1).

Conditional independence does not imply marginal independence, and conversely (Simpson's paradox). Both hold when one of the stronger structures applies (Figure 1).

8.2.2 Homogeneous association and three-factor interaction

Definition (Homogeneous association and the saturated model)

The model of *homogeneous association* (no three-factor interaction) is

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}, \tag{8.5}$$

symbol (XY, XZ, YZ); equivalently $\pi_{ijk} = \psi_{ij}\phi_{jk}\omega_{ik}$. The general (saturated) model adds λ_{ijk}^{XYZ}, symbol (XYZ).

The saturated model has

$$1+(I-1)+(J-1)+(K-1)+(I-1)(J-1)+(I-1)(K-1)+(J-1)(K-1)+(I-1)(J-1)(K-1) = IJK$$

parameters, one per cell (expand $IJK = [(I-1)+1][(J-1)+1][(K-1)+1]$). Unlike the models of Theorem 8.6, (8.5) has no closed-form expression of ψ, ϕ, ω in terms of margins.

Model symbol	Association terms	Interpretation
(X, Y, Z)	none	mutual independence
(XY, Z)	λ^{XY}	Z jointly independent of X, Y
(XY, YZ)	$\lambda^{XY}, \lambda^{YZ}$	$X \perp Z \mid Y$
(XY, XZ, YZ)	all two-factor	homogeneous association
(XYZ)	all, incl. λ^{XYZ}	saturated

The symbol (XZ, YZ) corresponds to the formula `X*Z + Y*Z` in R.

8.2.3 Interpreting model parameters

At level k of Z the XY conditional association is described by the $(I-1)(J-1)$ local odds ratios

$$\theta_{ij(k)} = \frac{\mu_{ijk}\,\mu_{i+1,j+1,k}}{\mu_{i,j+1,k}\,\mu_{i+1,j,k}}, \tag{8.6}$$

and similarly $\theta_{i(j)k}$ for XZ and $\theta_{(i)jk}$ for YZ. Conditional independence of X, Y given Z means all $\theta_{ij(k)} = 1$.

Theorem 8.7 (Homogeneous association)

(a) Under (XY, XZ, YZ),

$$\log \theta_{ij(k)} = \lambda_{ij}^{XY} + \lambda_{i+1,j+1}^{XY} - \lambda_{i,j+1}^{XY} - \lambda_{i+1,j}^{XY} \quad \text{for all } k, \tag{8.7}$$

and similarly for XZ and YZ odds ratios.
(b) Conversely, a positive table satisfies (XY, XZ, YZ) if and only if $\theta_{ij(1)} = \cdots = \theta_{ij(K)}$ for all i, j. Equivalently, iff the XZ odds ratios are the same at every j, or iff the YZ odds ratios are the same at every i.

Proof. [Supplement]

Step 1. For (a), substitute (8.5) into $\log \theta_{ij(k)}$. Each of $\lambda, \lambda^X, \lambda^Y, \lambda_k^Z, \lambda_{ik}^{XZ}, \lambda_{i+1,k}^{XZ}, \lambda_{jk}^{YZ}, \lambda_{j+1,k}^{YZ}$ appears once with each sign and cancels; only the XY contrast remains, and it does not involve k.

Step 2. For (b), suppose $\theta_{ij(k)} = \theta_{ij}$ does not depend on k. Let $h_{ijk} = \log \mu_{ijk}$. Telescoping the local log odds ratios over the rectangle $[1, i] \times [1, j]$,

$$h_{ijk} - h_{1jk} - h_{i1k} + h_{11k} = \sum_{a<i} \sum_{b<j} \log \theta_{ab} =: g(i, j),$$

which is free of k. Therefore

$$h_{ijk} = g(i, j) + [h_{1jk} - h_{11k}] + h_{i1k},$$

a sum of a function of (i, j), a function of (j, k) and a function of (i, k). By Lemma 8.5 the table satisfies (XY, XZ, YZ).

Step 3. The symmetric statements follow by the same argument with the roles of the variables permuted; since each is equivalent to (XY, XZ, YZ), they are equivalent to one another. ■

The three-factor term describes how an odds ratio changes across the third variable. For a $2 \times 2 \times 2$ table, direct substitution of the saturated model gives

$$\log \frac{\theta_{11(1)}}{\theta_{11(2)}} = (\lambda_{111}^{XYZ} + \lambda_{221}^{XYZ} - \lambda_{121}^{XYZ} - \lambda_{211}^{XYZ}) - (\lambda_{112}^{XYZ} + \lambda_{222}^{XYZ} - \lambda_{122}^{XYZ} - \lambda_{212}^{XYZ}),$$

which equals λ_{111}^{XYZ} when second-category parameters are set to 0. When X, Y are binary and second-level parameters are zero, $\log \theta_{11(k)} = \lambda_{11}^{XY}$.

8.2.4 Alcohol, cigarette, and marijuana use example

Example 8.8 (Alcohol, cigarette, marijuana use)

A 1992 survey of 2276 high-school seniors near Dayton, Ohio, recorded whether they had ever used alcohol (A), cigarettes (C) and marijuana (M).

		Observed: M		Fit of (AC, AM, CM)	
A	C	yes	no	yes	no
yes	yes	911	538	910.4	538.6
yes	no	44	456	44.6	455.4
no	yes	3	43	3.6	42.4
no	no	2	279	1.4	279.6

Estimated odds ratios under several models:

	Conditional			Marginal		
Model	AC	AM	CM	AC	AM	CM
(A, C, M)	1.0	1.0	1.0	1.0	1.0	1.0
(AC, M)	17.7	1.0	1.0	17.7	1.0	1.0
(AM, CM)	1.0	61.9	25.1	2.7	61.9	25.1
(AC, AM, CM)	7.8	19.8	17.3	17.7	61.9	25.1
(ACM), M=yes	13.8	24.3	17.5	17.7	61.9	25.1
(ACM), M=no	7.7	13.5	9.7			

A term absent from a model gives a conditional odds ratio 1.0; e.g. under (AM, CM),

$$\frac{909.24 \times 0.24}{45.76 \times 4.76} = \frac{438.84 \times 179.84}{555.16 \times 142.16} = 1.0,$$

yet the fitted marginal AC odds ratio is 2.7: conditional independence does not imply marginal independence. Under (AC, AM, CM) the AC conditional odds ratio is 7.8 at both levels of M, $= \exp(\hat{\lambda}_{11}^{AC} + \hat{\lambda}_{22}^{AC} - \hat{\lambda}_{12}^{AC} - \hat{\lambda}_{21}^{AC})$. Estimated odds ratios depend heavily on the model, so model selection matters.

8.3 Inference for loglinear models

Motivation. Once a loglinear model is fitted, we need to know whether it fits, which association terms are needed, and how large the associations are. The tools are the usual deviance tests and Wald/likelihood intervals for GLMs.

8.3.1 Chi-squared goodness-of-fit tests

$X^2 = \sum(n - \hat{\mu})^2/\hat{\mu}$ and $G^2 = 2\sum n \log(n/\hat{\mu})$ test model fit, with df = number of cells − number of nonredundant parameters. Model selection can also use AIC $= -2$(maximized log likelihood − number of parameters), equivalently minimizing $G^2 - 2\,\text{df}$.

Example 8.9 (Goodness of fit for the student survey)

Model	G^2	X^2	df	P-value
(A, C, M)	1286.0	1411.4	4	< 0.001
(A, CM)	534.2	505.6	3	< 0.001
(C, AM)	939.6	824.2	3	< 0.001
(M, AC)	843.8	704.9	3	< 0.001
(AC, AM)	497.4	443.8	2	< 0.001
(AC, CM)	92.0	80.8	2	< 0.001
(AM, CM)	187.8	177.6	2	< 0.001
(AC, AM, CM)	0.4	0.4	1	0.54
(ACM)	0.0	0.0	0	–

Only the homogeneous association model fits; it also minimizes AIC.

8.3.2 Inference about conditional associations

Tests about a term compare nested models. The likelihood-ratio statistic $-2(L_0 - L_1)$ equals $G^2(M_0) - G^2(M_1)$, the difference of deviances. For XY conditional independence within (XY, XZ, YZ), $H_0 : \lambda_{ij}^{XY} = 0$ for all $(I-1)(J-1)$ parameters, and the statistic is

$$G^2[(XZ, YZ) \mid (XY, XZ, YZ)] = G^2(XZ, YZ) - G^2(XY, XZ, YZ), \qquad \text{df} = (I-1)(J-1).$$

This is a model-based alternative to the generalized CMH test.

Example 8.10 (Conditional associations in the student survey)

For AC conditional independence, $G^2[(AM, CM) \mid (AC, AM, CM)] = 187.8 - 0.4 = 187.4$, df $= 1$, $P < 0.001$. The other two pairwise terms are likewise needed (LR statistics 91.6 for AM, 497.0 for CM). With baseline coding (last category $= 0$), (AC, AM, CM) gives

$$\hat{\lambda}_{11}^{AC} = 2.054 \text{ (SE} = 0.174), \quad \hat{\lambda}_{11}^{AM} = 2.986 \text{ (0.465)}, \quad \hat{\lambda}_{11}^{CM} = 2.848 \text{ (0.164)}.$$

For binary variables these are conditional log odds ratios. The 95% Wald interval for the AC conditional odds ratio is $\exp(2.054 \pm 1.96 \times 0.174) = (5.5, 11.0)$; for AM it is (8.0, 49.2) and for CM (12.5, 23.8). All associations are strong, and the marginal ones are stronger still: controlling for one response moderates the association between the other two.

With large samples, significance alone is not informative; confidence intervals show whether associations are practically important.

8.4 Loglinear models for higher dimensions

Motivation. The three-way ideas extend directly to four or more variables. The difficulties are the large number of possible terms, which makes model selection harder, and the large number of cells, which can make tables sparse.

8.4.1 Four-way contingency tables

For variables W, X, Y, Z, the model with all two-factor terms,

$$\log \mu_{hijk} = \lambda + \lambda_h^W + \lambda_i^X + \lambda_j^Y + \lambda_k^Z$$

$$+\lambda_{hi}^{WX}+\lambda_{hj}^{WY}+\lambda_{hk}^{WZ}+\lambda_{ij}^{XY}+\lambda_{ik}^{XZ}+\lambda_{jk}^{YZ},$$

symbol (WX, WY, WZ, XY, XZ, YZ), has each pair conditionally dependent with the same odds ratios at every combination of levels of the other two variables. Removing a two-factor term means conditional independence of that pair given the other two variables. Under (WXY, WZ, XZ, YZ), the WX, WY, XY associations may vary across levels of the remaining variable of the triple, while every association involving Z is homogeneous. The saturated model adds all three-factor terms and λ^{WXYZ}.

Proposition 8.11 (Odds ratios in models without a variable in higher-order terms)

If a variable V appears in no term of order three or more, then for every other variable U the conditional UV odds ratios are the same at every combination of levels of the remaining variables, and they equal the exponentiated contrast of λ^{UV}.

Proof. [Supplement] Fix levels of the remaining variables and form a UV odds ratio $\mu_{ab\cdot}\mu_{cd\cdot}/(\mu_{ad\cdot}\mu_{cb\cdot})$ (U levels a, c; V levels b, d). In the log, any term not involving V appears with U at level a once with each sign (in μ_{ab} and μ_{ad}), and likewise at level c, so it cancels. Any term not involving U cancels the same way. The only terms involving both U and V are, by hypothesis, the two-factor term λ^{UV}, which does not depend on the other variables. Hence the log odds ratio is $\lambda_{ab}^{UV}+\lambda_{cd}^{UV}-\lambda_{ad}^{UV}-\lambda_{cb}^{UV}$ at every level combination. ■

8.4.2 Automobile accident example

Example 8.12 (Automobile accidents and seat belts)

In 1991, 68,694 passengers in auto accidents in Maine were classified by gender (G), location (L: urban, rural), seat-belt use (S) and injury (I).

Gender	Location	Seat belt	Injury no	Injury yes
Female	Urban	No	7287	996
		Yes	11587	759
	Rural	No	3246	973
		Yes	6134	757
Male	Urban	No	10381	812
		Yes	10969	380
	Rural	No	6123	1084
		Yes	6693	513

Goodness of fit:

Model	G^2	df	P
(G, I, L, S)	2792.8	11	< 0.0001
(GI, GL, GS, IL, IS, LS)	23.4	5	< 0.001
(GIL, GIS, GLS, ILS)	1.3	1	0.25
(GIL, GS, IS, LS)	18.6	4	0.001
(GIS, GL, IL, LS)	22.8	4	< 0.001
(GLS, GI, IL, IS)	7.5	4	0.11
(ILS, GI, GL, GS)	20.6	4	< 0.001

Estimated conditional odds ratios:

Odds ratio	(GI, GL, GS, IL, IS, LS)	(GLS, GI, IL, IS)
GI	0.58	0.58
IL	2.13	2.13
IS	0.44	0.44
GL (S=no / yes)	1.23 / 1.23	1.33 / 1.17
GS (L=urban / rural)	0.63 / 0.63	0.66 / 0.58
LS (G=female / male)	1.09 / 1.09	1.17 / 1.03

Under the pairwise model the IS conditional log odds ratio is -0.814 (SE 0.028), so the 95% interval for the odds ratio is $\exp(-0.814 \pm 1.96 \times 0.028) = (0.42, 0.47)$: wearing a seat belt more than halves the odds of injury at each gender–location combination. Injury is more likely in rural accidents and for females; males have 0.63 times the odds of females of wearing seat belts. In (GLS, GI, IL, IS) the variable I is in no three-factor term, so (Proposition 8.11) its odds ratios with G, L, S are homogeneous; the GL, GS, LS odds ratios vary with the third variable, e.g. $0.66 = (7273.2 \times 10{,}959.2)/(11{,}632.6 \times 10{,}358.9)$ for urban accidents.

8.4.3 Large samples and statistical versus practical significance

(GLS, GI, IL, IS) fits significantly better than the pairwise model: $23.4 - 7.5 = 15.9$, df $= 1$, $P = 0.0001$. Yet the fitted GL, GS, LS odds ratios are similar at both levels of the third variable, so the three-factor interaction is weak; the significance reflects the huge n. With large samples, focus on estimation; the simpler pairwise model is adequate for most purposes.

8.4.4 Dissimilarity index

Definition (Dissimilarity index)

For cell counts $n_i = np_i$ and fitted values $\hat{\mu}_i = n\hat{\pi}_i$,

$$\hat{\Delta} = \frac{\sum_i |n_i - \hat{\mu}_i|}{2n} = \frac{1}{2}\sum_i |p_i - \hat{\pi}_i|.$$

It estimates the population index $\Delta = \frac{1}{2}\sum |\pi_i - \pi_i^{\text{model}}|$.

Proposition 8.13 (Meaning of $\hat{\Delta}$)

If $\sum \hat{\mu}_i = n$ (true for any loglinear model with an intercept), then $0 \le \hat{\Delta} \le 1$, and $n\hat{\Delta}$ is the minimum number of observations that must be moved to other cells to make the observed table equal to the fitted one.

Proof. [Supplement]

Step 1. $\sum |p_i - \hat{\pi}_i| \le \sum p_i + \sum \hat{\pi}_i = 2$, so $\hat{\Delta} \le 1$; clearly $\hat{\Delta} \ge 0$.

Step 2. Let $d_i = n_i - \hat{\mu}_i$. Since $\sum d_i = 0$, $\sum d_i^+ = \sum d_i^- = \frac{1}{2}\sum |d_i| = n\hat{\Delta}$.

Step 3. Moving one observation changes $\sum d_i^+$ by at most 1, so at least $\sum d_i^+$ moves are necessary. Moving d_i^+ observations out of each cell with surplus into cells with deficits (total deficit $\sum d_i^-$, the same amount) is sufficient (allowing fractional "observations" since $\hat{\mu}$ need not be integer). Hence the minimum is $n\hat{\Delta}$. ■

Values $\hat{\Delta} < 0.02$ or 0.03 indicate that the data follow the model closely in practical terms even if

the model is rejected; $\hat{\Delta}$ tends to overestimate Δ when Δ is near 0, especially for small n. For the accident data, $\hat{\Delta} = 0.008$ for the pairwise model and 0.003 for (GLS, GI, IL, IS): moving less than 1% of the data yields a perfect fit.

8.5 Loglinear–logit model connection

Motivation. Loglinear models treat variables symmetrically, logit models single out a response. Forming a logit from a loglinear model helps interpret it, and every logit model with categorical predictors is equivalent to a loglinear model that contains the full interaction among the predictors.

8.5.1 Using logit models to interpret loglinear models

Proposition 8.14 (Logit form of (XY, XZ, YZ))

If Y is binary and (XY, XZ, YZ) holds, then

$$\text{logit}\, P(Y = 1 \mid X = i, Z = k) = \alpha + \beta_i^X + \beta_k^Z, \tag{8.8}$$

with $\alpha = \lambda_1^Y - \lambda_2^Y$, $\beta_i^X = \lambda_{i1}^{XY} - \lambda_{i2}^{XY}$, $\beta_k^Z = \lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}$.

Proof. The logit is $\log(\mu_{i1k}/\mu_{i2k})$. Subtracting (8.5) at $j = 2$ from (8.5) at $j = 1$, the terms $\lambda, \lambda_i^X, \lambda_k^Z, \lambda_{ik}^{XZ}$ cancel, leaving

$$(\lambda_1^Y - \lambda_2^Y) + (\lambda_{i1}^{XY} - \lambda_{i2}^{XY}) + (\lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}).$$

■

The λ^{XZ} terms, which describe association among explanatory variables, cancel: the logit model $(X + Z)$ says nothing about it.

8.5.2 Auto accident example revisited

Example 8.15 (Injury as a response)

For (GLS, GI, IL, IS), forming the logit on injury cancels every term without the index i of I, including λ^{GLS}, giving the logit model $(G + L + S)$:

$$\text{logit}\, P(I = 1 \mid G = g, L = l, S = s) = \alpha + \beta_g^G + \beta_l^L + \beta_s^S, \qquad \beta_s^S = \lambda_{1s}^{IS} - \lambda_{2s}^{IS}.$$

Fitted values, G^2, residual df and standardized residuals are identical for the two models. The seat-belt log odds ratio is $\hat{\beta}_1^S - \hat{\beta}_2^S = -0.817$ in the logit model and $\hat{\lambda}_{11}^{IS} + \hat{\lambda}_{22}^{IS} - \hat{\lambda}_{12}^{IS} - \hat{\lambda}_{21}^{IS} = -0.817$ in the loglinear model. The loglinear model treats the 16 counts as Poisson; the logit model treats the GLS margin as fixed and the 8 counts $\{n_{g1ls}\}$ as binomial.

8.5.3 Correspondence between loglinear and logit models

Forming the logit on Y in (XY, YZ) also gives (8.8), but (XY, YZ) is *not* equivalent to the logit model: it additionally forces $X \perp Z \mid Y$. The equivalent loglinear model must contain the most general interaction among the explanatory variables.

Theorem 8.16 (Loglinear–logit equivalence)

Let Y be the response and $\mathbf{x}$ index the combinations of the explanatory variables. Consider a loglinear model containing the full interaction term among the explanatory variables, $\log \mu_{\mathbf{x}j} = \gamma_{\mathbf{x}} + \eta_{\mathbf{x}j}(\boldsymbol{\beta})$, where $\gamma_{\mathbf{x}}$ is unrestricted and $\eta_{\mathbf{x}j}$ collects the terms involving Y (with $\eta_{\mathbf{x}J} = 0$). Then the ML fitted values under Poisson sampling coincide with those of the baseline-category logit model $\log(\pi_{j|\mathbf{x}}/\pi_{J|\mathbf{x}}) = \eta_{\mathbf{x}j}(\boldsymbol{\beta})$ fitted with the margins $n_{\mathbf{x}+}$ fixed, and the ML estimates of $\boldsymbol{\beta}$ agree.

Proof. [Supplement]

Step 1. Reparametrize: $\mu_{\mathbf{x}+} = e^{\gamma_{\mathbf{x}}} \sum_j e^{\eta_{\mathbf{x}j}}$ and $\pi_{j|\mathbf{x}} = e^{\eta_{\mathbf{x}j}} / \sum_h e^{\eta_{\mathbf{x}h}}$, so $\mu_{\mathbf{x}j} = \mu_{\mathbf{x}+}\pi_{j|\mathbf{x}}$. Since $\gamma_{\mathbf{x}}$ is free, $(\{\mu_{\mathbf{x}+}\}, \boldsymbol{\beta})$ ranges over $(0,\infty)^{\#\mathbf{x}} \times \{\text{all } \boldsymbol{\beta}\}$, a product set (variation independence).

Step 2. The Poisson log likelihood splits:

$$\sum_{\mathbf{x},j}(n_{\mathbf{x}j}\log\mu_{\mathbf{x}j} - \mu_{\mathbf{x}j}) = \sum_{\mathbf{x}}(n_{\mathbf{x}+}\log\mu_{\mathbf{x}+} - \mu_{\mathbf{x}+}) + \sum_{\mathbf{x},j} n_{\mathbf{x}j}\log\pi_{j|\mathbf{x}}(\boldsymbol{\beta}).$$

Step 3. The two parts depend on separate parameters, so they are maximized separately: $\hat{\mu}_{\mathbf{x}+} = n_{\mathbf{x}+}$, and $\hat{\boldsymbol{\beta}}$ maximizes the second part, which is exactly the product-multinomial (binomial if $J = 2$) log likelihood of the logit model. Hence $\hat{\mu}_{\mathbf{x}j} = n_{\mathbf{x}+}\hat{\pi}_{j|\mathbf{x}}$ in both models. ■

Loglinear symbol	Logit model	Logit symbol
(Y, XZ)	α	$(-)$
(XY, XZ)	$\alpha + \beta_i^X$	(X)
(YZ, XZ)	$\alpha + \beta_k^Z$	(Z)
(XY, YZ, XZ)	$\alpha + \beta_i^X + \beta_k^Z$	$(X + Z)$
(XYZ)	$\alpha + \beta_i^X + \beta_k^Z + \beta_{ik}^{XZ}$	$(X*Z)$

With several response variables the loglinear approach is more general; with one response, logit models are more natural.

8.5.4 Generalized loglinear model

Write the counts as a vector $\mathbf{n} = (n_1, \ldots, n_N)'$ with means $\boldsymbol{\mu}$. An ordinary loglinear model is

$$\log \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}. \tag{8.9}$$

For instance, independence in a 2×2 table with $\lambda_2^X = \lambda_2^Y = 0$ is

$$\begin{pmatrix} \log\mu_{11} \\ \log\mu_{12} \\ \log\mu_{21} \\ \log\mu_{22} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ \lambda_1^X \\ \lambda_1^Y \end{pmatrix}.$$

The *generalized loglinear model* is $\mathbf{C}\log(\mathbf{A}\boldsymbol{\mu}) = \mathbf{X}\boldsymbol{\beta}$. With $\mathbf{A} = \mathbf{I}_4$, $\mathbf{C} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$, $\mathbf{X} = (1,1)'$ and $\boldsymbol{\beta} = \alpha$, it states that the logits in both rows equal α, i.e. independence again. The class includes logit models and, later, models for marginal distributions.

8.6 Loglinear model fitting: likelihood equations and asymptotic distributions*

Motivation. Why are fitted values of loglinear models obtained by matching marginal tables? Because the Poisson model is an exponential family whose sufficient statistics are those margins. This section derives the sufficient statistics, likelihood equations, uniqueness, closed-form fits, degrees of freedom and the asymptotic covariance matrix.

Derivations use Poisson sampling, which needs no constraint on parameters; Theorem 8.23 transfers them to multinomial sampling.

8.6.1 Minimal sufficient statistics

The kernel of the Poisson log likelihood for a three-way table is

$$L(\boldsymbol{\mu}) = \sum_{i,j,k} n_{ijk} \log \mu_{ijk} - \sum_{i,j,k} \mu_{ijk}. \tag{8.10}$$

Substituting the saturated model,

$$\begin{aligned} L = n\lambda + \sum_i n_{i++}\lambda_i^X + \sum_j n_{+j+}\lambda_j^Y + \sum_k n_{++k}\lambda_k^Z + \sum_{i,j} n_{ij+}\lambda_{ij}^{XY} + \sum_{i,k} n_{i+k}\lambda_{ik}^{XZ} \\ + \sum_{j,k} n_{+jk}\lambda_{jk}^{YZ} + \sum_{i,j,k} n_{ijk}\lambda_{ijk}^{XYZ} - \sum_{i,j,k} \exp(\lambda + \lambda_i^X + \cdots + \lambda_{ijk}^{XYZ}). \end{aligned} \tag{8.11}$$

Theorem 8.17 (Sufficient statistics are marginal tables)

For a hierarchical loglinear model, the minimal sufficient statistics are the marginal tables corresponding to the terms in the model symbol: e.g. $\{n_{i++}\}, \{n_{+j+}\}, \{n_{++k}\}$ for (X, Y, Z); $\{n_{ij+}\}, \{n_{++k}\}$ for (XY, Z); $\{n_{ij+}\}, \{n_{+jk}\}$ for (XY, YZ); $\{n_{ij+}\}, \{n_{i+k}\}, \{n_{+jk}\}$ for (XY, XZ, YZ).

Proof.

Step 1. Setting the absent parameters to zero in (8.11), the joint distribution is an exponential family whose canonical statistics are the coefficients of the remaining parameters, i.e. the marginal tables of all terms present. By the factorization theorem these are sufficient; minimality holds because, after removing redundant parameters, the family is full rank.

Step 2. A lower-order margin is a sum of a higher-order one (e.g. $n_{i++} = \sum_j n_{ij+}$), so only the margins of the highest-order terms (the generators) are needed. ■

For the saturated model the full table is sufficient: no data reduction.

8.6.2 Likelihood equations for loglinear models

In the matrix form (8.9), $\log \mu_i = \sum_j x_{ij}\beta_j$ and

$$L(\boldsymbol{\beta}) = \sum_i n_i \Big(\sum_j x_{ij}\beta_j \Big) - \sum_i \exp\Big(\sum_j x_{ij}\beta_j \Big). \tag{8.12}$$

Theorem 8.18 (Likelihood equations match margins)

(a) The likelihood equations are

$$\mathbf{X}'\mathbf{n} = \mathbf{X}'\hat{\boldsymbol{\mu}}: \tag{8.13}$$

sufficient statistics equal their fitted (expected) values.
(b) For a hierarchical model, (8.13) is equivalent to equality of the observed and fitted marginal tables in the model symbol. For (XZ, YZ):

$$\hat{\mu}_{i+k} = n_{i+k} \text{ for all } i,k, \qquad \hat{\mu}_{+jk} = n_{+jk} \text{ for all } j,k. \tag{8.14}$$

Proof.

Step 1. Since $\partial \exp(\sum_j x_{ij}\beta_j)/\partial\beta_j = x_{ij}\mu_i$,

$$\frac{\partial L}{\partial \beta_j} = \sum_i n_i x_{ij} - \sum_i \mu_i x_{ij}, \qquad j = 1, \ldots, p.$$

Setting these to 0 gives (8.13).

Step 2. (8.13) says $\mathbf{v}'\mathbf{n} = \mathbf{v}'\hat{\boldsymbol{\mu}}$ for every $\mathbf{v}$ in the column space of $\mathbf{X}$. By Lemma 8.5, this column space is the set of functions $\sum_r g_r(\mathbf{x}_{C_r})$, which is spanned by the indicator vectors $\mathbf{v} = \mathbb{1}\{\mathbf{x}_{C_r} = \mathbf{c}\}$ of the cells of each generator's margin. For such $\mathbf{v}$, $\mathbf{v}'\mathbf{n}$ is the marginal count at $\mathbf{c}$. Hence (8.13) is equivalent to equality of all generator margins. For (XZ, YZ) this is (8.14). ■

Thus fitted values are smoothed versions of the data that agree with them on certain margins but whose associations follow the model.

8.6.3 Birch's results for loglinear models

Theorem 8.19 (Birch)

There is at most one $\hat{\boldsymbol{\mu}}$ that both satisfies the loglinear model and matches the data in the minimal sufficient statistics, and if it exists it is the ML estimate. Conversely, any ML estimate has these two properties.

Proof. [Supplement]

Step 1. Write $\boldsymbol{\eta} = \log \boldsymbol{\mu} \in \mathcal{V} = \text{col}(\mathbf{X})$. As a function of $\boldsymbol{\eta}$, $L = \mathbf{n}'\boldsymbol{\eta} - \sum_i e^{\eta_i}$ is strictly concave on $\mathbb{R}^N$: its Hessian is $-\text{diag}(e^{\eta_i})$, negative definite. Its restriction to the subspace $\mathcal{V}$ is still strictly concave.

Step 2. A point $\boldsymbol{\eta} \in \mathcal{V}$ is stationary for $L|_{\mathcal{V}}$ iff the gradient $\mathbf{n} - \boldsymbol{\mu}$ is orthogonal to $\mathcal{V}$, i.e. $\mathbf{X}'(\mathbf{n} - \boldsymbol{\mu}) = \mathbf{0}$, which is margin matching by Theorem 8.18.

Step 3. For a differentiable concave function, a stationary point is a global maximum; for a strictly concave one, there is at most one maximizer. Hence a $\boldsymbol{\mu}$ in the model that matches the margins is the unique MLE, and conversely the MLE (an interior maximizer) is stationary, so it matches the margins. ■

For example, for independence in a two-way table, $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$ satisfies the model and has $\hat{\mu}_{i+} = n_{i+}$, $\hat{\mu}_{+j} = n_{+j}$; by Birch it is the MLE.

8.6.4 Direct versus iterative calculation of fitted values

Theorem 8.20 (Closed-form (direct) estimates)

The ML fitted values for three-way models other than (XY, XZ, YZ) are:

Model	Probabilistic form	Fitted value $\hat{\mu}_{ijk}$
(X, Y, Z)	$\pi_{i++}\pi_{+j+}\pi_{++k}$	$n_{i++}n_{+j+}n_{++k}/n^2$
(XY, Z)	$\pi_{ij+}\pi_{++k}$	$n_{ij+}n_{++k}/n$
(XY, XZ)	$\pi_{ij+}\pi_{i+k}/\pi_{i++}$	$n_{ij+}n_{i+k}/n_{i++}$
(XY, XZ, YZ)	$\psi_{ij}\phi_{jk}\omega_{ik}$	iterative (Section 8.7)
(XYZ)	none	n_{ijk}

(other models by symmetry, e.g. (XZ, YZ): $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$).

Proof. Take (XZ, YZ); the others are identical in structure.

Step 1. By Theorem 8.6, under the model $\mu_{ijk} = \mu_{i+k}\mu_{+jk}/\mu_{++k}$.

Step 2. Define $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$. It has the product form $A_{ik}B_{jk}$, so it satisfies the model.

Step 3. Summing over j: $\hat{\mu}_{i+k} = n_{i+k}n_{++k}/n_{++k} = n_{i+k}$; summing over i: $\hat{\mu}_{+jk} = n_{+jk}$. Thus it matches the sufficient margins (8.14).

Step 4. By Birch's theorem it is the MLE. For (X, Y, Z), (XY, Z), (XY, XZ) the displayed fits likewise have the factorized form and reproduce the sufficient margins (e.g. for (XY, XZ): $\sum_k n_{ij+}n_{i+k}/n_{i++} = n_{ij+}$ and $\sum_j n_{ij+}n_{i+k}/n_{i++} = n_{i+k}$). For (XYZ), $\hat{\mu} = \mathbf{n}$ trivially. ■

Models with closed-form estimates are called *direct* (or *decomposable*: fitted values are products and ratios of fitted margins). Unsaturated models containing all two-factor terms, such as (XY, XZ, YZ), have no direct estimates. Software uses iterative methods for all models anyway.

8.6.5 Chi-squared goodness-of-fit tests

For a model with an intercept, the deviance equals G^2; for a fixed number of cells and large expected counts, G^2 and X^2 are approximately χ^2 with df = (number of parameters in general) − (number under the model).

Proposition 8.21 (Residual df)

For a hierarchical model, residual df = number of cells − number of nonredundant parameters, which equals the sum, over all terms excluded from the model, of the products $\prod_{V \in \text{term}}(\#\text{levels of } V - 1)$. For three-way tables:

Model	df	Model	df
(X, Y, Z)	$IJK - I - J - K + 2$	(XY, YZ)	$J(I-1)(K-1)$
(XY, Z)	$(K-1)(IJ-1)$	(XZ, YZ)	$K(I-1)(J-1)$
(XZ, Y)	$(J-1)(IK-1)$	(XY, XZ)	$I(J-1)(K-1)$
(YZ, X)	$(I-1)(JK-1)$	(XY, XZ, YZ)	$(I-1)(J-1)(K-1)$

The df are the same for multinomial and Poisson sampling.

Proof. [Supplement]

Step 1. Expanding $IJK = \prod_V[(L_V - 1) + 1]$ writes the number of cells as the sum, over all subsets T of variables, of $\prod_{V \in T}(L_V - 1)$, which is the number of nonredundant parameters of the term λ^T

under baseline coding (the empty set gives the intercept). The saturated model has all terms, so the residual df of a model is the sum over excluded terms.

Step 2. Examples: (XY, XZ, YZ) excludes only XYZ: df $= (I-1)(J-1)(K-1)$. (XZ, YZ) excludes XY and XYZ: $(I-1)(J-1)+(I-1)(J-1)(K-1) = K(I-1)(J-1)$. (X, Y, Z) excludes all interaction terms: $IJK - [1 + (I-1) + (J-1) + (K-1)] = IJK - I - J - K + 2$. (XY, Z): $IJK - [1 + (I-1) + (J-1) + (K-1) + (I-1)(J-1)] = IJK - IJ - K + 1 = (K-1)(IJ-1)$. The rest follow by symmetry.

Step 3. Under multinomial sampling there are $IJK - 1$ free probabilities and the model loses λ (fixed by $\sum \pi = 1$), so both counts drop by one and the df are unchanged. ■

8.6.6 Covariance matrix of ML parameter estimators

Theorem 8.22 (Information and asymptotic covariance)

For Poisson loglinear models with full-rank $\mathbf{X}$, the observed and expected information coincide:

$$\mathcal{I}(\boldsymbol{\beta}) = -\frac{\partial^2 L}{\partial \boldsymbol{\beta}\, \partial \boldsymbol{\beta}'} = \mathbf{X}' \operatorname{diag}(\boldsymbol{\mu}) \mathbf{X},$$

and for a fixed number of cells as $n \to \infty$, $\hat{\boldsymbol{\beta}}$ is asymptotically normal with

$$\widehat{\operatorname{cov}}(\hat{\boldsymbol{\beta}}) = \left[\mathbf{X}' \operatorname{diag}(\hat{\boldsymbol{\mu}}) \mathbf{X}\right]^{-1}. \tag{8.15}$$

Proof. Differentiate $\partial L / \partial \beta_j = \sum_i (n_i - \mu_i) x_{ij}$ once more: $\partial \mu_i / \partial \beta_k = \mu_i x_{ik}$, so

$$\frac{\partial^2 L}{\partial \beta_j \partial \beta_k} = -\sum_i \mu_i x_{ij} x_{ik}.$$

This is free of $\mathbf{n}$ (canonical link), so the observed and expected information both equal $\mathbf{X}' \operatorname{diag}(\boldsymbol{\mu}) \mathbf{X}$. The asymptotic normality with covariance equal to the inverse information is the standard ML result for GLMs; substituting $\hat{\boldsymbol{\mu}}$ gives (8.15). ■

8.6.7 Connection between multinomial and Poisson loglinear models

Theorem 8.23 (Poisson and multinomial give the same estimates)

Write the Poisson model as $\log \mu_i = \alpha + \mathbf{x}_i' \boldsymbol{\beta}$ (α the intercept). Then the Poisson log likelihood decomposes as

$$L(\alpha, \boldsymbol{\beta}) = \Big[\sum_i n_i \log \pi_i(\boldsymbol{\beta})\Big] + [n \log \tau - \tau], \qquad \pi_i(\boldsymbol{\beta}) = \frac{e^{\mathbf{x}_i' \boldsymbol{\beta}}}{\sum_h e^{\mathbf{x}_h' \boldsymbol{\beta}}}, \tag{8.16}$$

where $\tau = \sum_i \mu_i$. The first bracket is the multinomial log likelihood given n, the second the Poisson log likelihood of n. Consequently $\hat{\boldsymbol{\beta}}$ and its asymptotic covariance are the same under Poisson and multinomial sampling.

Proof.

Step 1. $L = \sum n_i \log \mu_i - \sum \mu_i = \sum n_i (\alpha + \mathbf{x}_i' \boldsymbol{\beta}) - \tau = n\alpha + \sum n_i \mathbf{x}_i' \boldsymbol{\beta} - \tau$.

Step 2. Since $\tau = e^{\alpha} \sum_h e^{\mathbf{x}_h' \boldsymbol{\beta}}$, $\alpha = \log \tau - \log \sum_h e^{\mathbf{x}_h' \boldsymbol{\beta}}$. Substituting,

$$L = \sum_i n_i \mathbf{x}_i' \boldsymbol{\beta} - n \log \sum_h e^{\mathbf{x}_h' \boldsymbol{\beta}} + n \log \tau - \tau = \sum_i n_i \log \pi_i(\boldsymbol{\beta}) + (n \log \tau - \tau).$$

Step 3. The map $(\alpha, \boldsymbol{\beta}) \mapsto (\tau, \boldsymbol{\beta})$ is one-to-one onto $(0, \infty) \times \mathbb{R}^p$, and $\boldsymbol{\beta}$ enters only the first bracket. Hence $\hat{\boldsymbol{\beta}}$ maximizes the multinomial log likelihood, $\hat{\tau} = n$, and the information matrix is block diagonal in $(\tau, \boldsymbol{\beta})$, so the $\boldsymbol{\beta}$ block of the inverse information is the same for both sampling schemes. ■

The same argument with several independent multinomial samples works as long as the model contains the term for the margin fixed by the design (e.g. λ^{XZ} when $\{n_{i+k}\}$ are fixed), which forces $\hat{\mu}_{i+k} = n_{i+k}$; this is Theorem 8.16. For a single multinomial sample, $\widehat{\mathrm{cov}}(\hat{\boldsymbol{\beta}}) = \{\mathbf{X}'[\mathrm{diag}(\hat{\boldsymbol{\mu}}) - \hat{\boldsymbol{\mu}}\hat{\boldsymbol{\mu}}'/n]\mathbf{X}\}^{-1}$, where $\mathbf{X}$ omits the intercept column; it equals (8.15) with the row and column for α deleted.

8.6.8 Distribution of probability estimators

Proposition 8.24 (Covariance of sample proportions)

Under multinomial$(n, \boldsymbol{\pi})$ sampling, $\mathrm{cov}(\mathbf{p}) = [\mathrm{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}']/n$.

Proof. Write $\mathbf{n} = \sum_{s=1}^{n} \mathbf{Y}_s$ with $\mathbf{Y}_s$ i.i.d. indicator vectors ($Y_{si} = 1$ if subject s is in cell i). Then $E(Y_{si}) = \pi_i$, $E(Y_{si}Y_{sh}) = \pi_i \mathbb{1}\{i = h\}$, so $\mathrm{cov}(\mathbf{Y}_s) = \mathrm{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}'$. Summing n independent copies and dividing by n^2 gives the result. ■

With I independent multinomial samples, $\mathrm{cov}(\mathbf{p})$ is block diagonal with blocks of this form. For an unsaturated model the delta method gives, for a single multinomial sample,

$$\widehat{\mathrm{cov}}(\hat{\boldsymbol{\pi}}) = \frac{1}{n}\hat{\mathbf{D}}\mathbf{X}(\mathbf{X}'\hat{\mathbf{D}}\mathbf{X})^{-1}\mathbf{X}'\hat{\mathbf{D}}, \qquad \hat{\mathbf{D}} = \mathrm{diag}(\hat{\boldsymbol{\pi}}) - \hat{\boldsymbol{\pi}}\hat{\boldsymbol{\pi}}'.$$

A practical advantage: a cell with $n_i = 0$ has sample proportion 0 with standard error 0, whereas a model fit usually gives a positive $\hat{\pi}_i$ and standard error.

8.6.9 Uniqueness of ML estimates

Theorem 8.25 (Existence and uniqueness when all $n_i > 0$)

If all $n_i > 0$ and $\mathbf{X}$ has full rank, the ML estimate $\hat{\boldsymbol{\beta}}$ exists, is unique, and is the unique solution of the likelihood equations.

Proof. [Supplement]

Step 1 (bounded and coercive). Each term $f_i(\eta_i) = n_i\eta_i - e^{\eta_i}$ satisfies $f_i \le n_i \log n_i - n_i$ and, because $n_i > 0$, $f_i(\eta_i) \to -\infty$ as $\eta_i \to \pm\infty$. Hence $L(\boldsymbol{\eta}) = \sum f_i(\eta_i) \to -\infty$ as $\|\boldsymbol{\eta}\| \to \infty$.

Step 2 (existence). Since $\mathbf{X}$ has full rank, $\|\mathbf{X}\boldsymbol{\beta}\| \ge c\|\boldsymbol{\beta}\|$ for some $c > 0$, so $L(\boldsymbol{\beta}) \to -\infty$ as $\|\boldsymbol{\beta}\| \to \infty$. Thus the superlevel set $\{L \ge L(\mathbf{0})\}$ is compact, and the continuous L attains its maximum there; with continuous first derivatives, the gradient vanishes at the maximum.

Step 3 (uniqueness). The negative Hessian $\mathbf{X}'\,\mathrm{diag}(\boldsymbol{\mu})\mathbf{X}$ is nonnegative definite, since $\mathbf{c}'\mathbf{X}'\,\mathrm{diag}(\boldsymbol{\mu})\mathbf{X}\mathbf{c} = \sum_i \mu_i(\sum_j x_{ij}c_j)^2 \ge 0$; it is nonsingular because $\mu_i > 0$ and $\mathbf{X}$ has full rank, hence positive definite. So L is strictly concave in $\boldsymbol{\beta}$ and has at most one stationary point, which is the maximum. ■

When some $n_i = 0$ the estimates may fail to exist; see Section 9.8 (Chapter 9).

8.7 Loglinear model fitting: iterative methods and their application*

Motivation. Models such as (XY, XZ, YZ) have no closed-form fit. Newton–Raphson and iterative proportional fitting (IPF) solve the likelihood equations; IPF is simple and transparent, and it is also the tool for raking a table to given margins.

8.7.1 Newton–Raphson method

Proposition 8.26 (Newton–Raphson as iteratively reweighted least squares)

For the Poisson loglinear model, Newton–Raphson iterates

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \left[\mathbf{X}' \operatorname{diag}(\boldsymbol{\mu}^{(t)})\mathbf{X}\right]^{-1}\mathbf{X}'(\mathbf{n} - \boldsymbol{\mu}^{(t)}), \qquad \boldsymbol{\mu}^{(t)} = \exp(\mathbf{X}\boldsymbol{\beta}^{(t)}).$$

Equivalently $\boldsymbol{\beta}^{(t+1)} = (\mathbf{X}'\mathbf{V}^{(t)}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{(t)}\mathbf{z}^{(t)}$ with $\mathbf{V}^{(t)} = \operatorname{diag}(\boldsymbol{\mu}^{(t)})$ and working response $z_i^{(t)} = \log \mu_i^{(t)} + (n_i - \mu_i^{(t)})/\mu_i^{(t)}$: weighted least squares for $\mathbf{z} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with $\operatorname{Var}(\epsilon_i) = 1/\mu_i^{(t)}$.

Proof.

Step 1. Newton–Raphson is $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - (\mathbf{H}^{(t)})^{-1}\mathbf{u}^{(t)}$ with score $u_j = \sum_i (n_i - \mu_i)x_{ij}$ and Hessian $h_{jk} = -\sum_i \mu_i x_{ij} x_{ik}$ (Theorem 8.22). In matrix form $\mathbf{u} = \mathbf{X}'(\mathbf{n} - \boldsymbol{\mu})$, $\mathbf{H} = -\mathbf{X}'\mathbf{V}\mathbf{X}$, giving the first formula.

Step 2. Since $\mathbf{X}\boldsymbol{\beta}^{(t)} = \log \boldsymbol{\mu}^{(t)}$, write $\mathbf{X}'\mathbf{V}\mathbf{X}\boldsymbol{\beta}^{(t)} + \mathbf{X}'(\mathbf{n} - \boldsymbol{\mu}^{(t)}) = \mathbf{X}'\mathbf{V}[\log \boldsymbol{\mu}^{(t)} + \mathbf{V}^{-1}(\mathbf{n} - \boldsymbol{\mu}^{(t)})] = \mathbf{X}'\mathbf{V}\mathbf{z}^{(t)}$, and multiply by $(\mathbf{X}'\mathbf{V}\mathbf{X})^{-1}$. ∎

One may start from $\boldsymbol{\mu}^{(0)} = \mathbf{n}$ (adding $\frac{1}{2}$ to empty cells); then $\boldsymbol{\beta}^{(1)}$ is the weighted least squares fit of $\log \mathbf{n} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$. Since L is concave, convergence is usually fast, and $-\mathbf{H}^{-1}$ at convergence is the estimated covariance (8.15), a by-product.

8.7.2 Iterative proportional fitting

Definition (IPF (Deming and Stephan, 1940))

(1) Start with $\{\mu^{(0)}\}$ satisfying a model no more complex than the one being fitted (e.g. all $\mu^{(0)} = 1$).
(2) Successively multiply by factors so as to match each marginal table among the minimal sufficient statistics.
(3) Repeat cycles until fitted and observed sufficient margins agree to the desired accuracy.

For (XY, XZ, YZ) a cycle has three steps:

$$\mu_{ijk}^{(1)} = \mu_{ijk}^{(0)} \frac{n_{ij+}}{\mu_{ij+}^{(0)}}, \qquad \mu_{ijk}^{(2)} = \mu_{ijk}^{(1)} \frac{n_{i+k}}{\mu_{i+k}^{(1)}}, \qquad \mu_{ijk}^{(3)} = \mu_{ijk}^{(2)} \frac{n_{+jk}}{\mu_{+jk}^{(2)}}.$$

After step 1 the XY margin matches; after step 2 the XZ margin matches but the XY margin generally no longer does; and so on.

Theorem 8.27 (IPF converges to the MLE)

Suppose all $n_{ijk} > 0$ and $\mu^{(0)} \equiv 1$. Then:
(a) every IPF iterate satisfies the model;

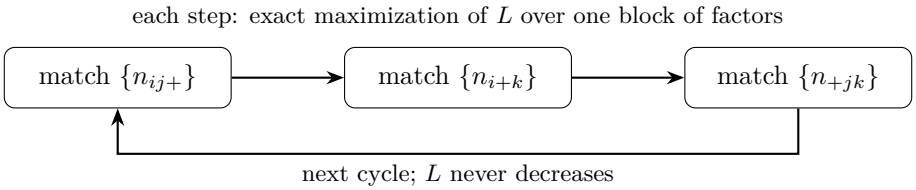


Figure 2: One IPF cycle for (XY, XZ, YZ).

(b) each step maximizes the Poisson log likelihood L over the block of factors it adjusts, so $L(\mu^{(t)})$ is nondecreasing (equivalently, G^2 comparing $\mathbf{n}$ with the current fit is nonincreasing); (c) the iterates converge to the unique $\hat{\boldsymbol{\mu}}$ that satisfies the model and matches all three margins, which by Birch's theorem is the MLE.
The same holds for any hierarchical model whose likelihood equations are margin matchings.

Proof. [Supplement]

Step 1 (a). Step 1 multiplies by $a_{ij} = n_{ij+}/\mu^{(0)}_{ij+}$, a function of (i, j) only. If $\mu^{(0)} = \psi_{ij}\phi_{jk}\omega_{ik}$, then $\mu^{(1)} = (a_{ij}\psi_{ij})\phi_{jk}\omega_{ik}$ has the same form; likewise for the other steps. Equivalently, every XY odds ratio at level k is multiplied by $a_{ij}a_{i+1,j+1}/(a_{i,j+1}a_{i+1,j})$, the same factor for all k, so homogeneity is preserved.

Step 2 (b). Fix $\mu^{(0)}$ and consider $\mu_{ijk} = c_{ij}\mu^{(0)}_{ijk}$ with free $c_{ij} > 0$. Then

$$L(c) = \sum_{i,j,k} n_{ijk} \log(c_{ij}\mu^{(0)}_{ijk}) - \sum_{i,j,k} c_{ij}\mu^{(0)}_{ijk} = \sum_{i,j}(n_{ij+} \log c_{ij} - c_{ij}\mu^{(0)}_{ij+}) + \text{const},$$

which is maximized uniquely at $c_{ij} = n_{ij+}/\mu^{(0)}_{ij+}$, exactly the IPF factor. Hence $L(\mu^{(1)}) \geq L(\mu^{(0)})$, and similarly for steps 2 and 3. Since $G^2 = 2[L(\mathbf{n}) - L(\boldsymbol{\mu})]$ when $\sum \mu = n$ (true after the first step), G^2 is nonincreasing.

Step 3 (bounded iterates). After the first step every iterate has total n, so $\mu_i \leq n$ and $L(\boldsymbol{\mu}) = \sum_h n_h \log \mu_h - n$. From $L(\mu^{(t)}) \geq L(\mu^{(1)})$ and $n_h \log \mu_h \leq n_h \log n$ for $h \neq i$,

$$n_i \log \mu^{(t)}_i \;\geq\; L(\mu^{(1)}) + n - \sum_{h \neq i} n_h \log n,$$

so, since $n_i > 0$, $\mu^{(t)}_i \geq \delta$ for a constant $\delta > 0$. The iterates lie in the compact set $K = \{\delta \leq \mu_i \leq n\}$, and the model points in K (those with $\log \boldsymbol{\mu}$ in the column space of $\mathbf{X}$) form a closed, hence compact, set.

Step 4 (limit points are fixed points). Let $\boldsymbol{\mu}^*$ be a limit of a subsequence $\mu^{(3s_r)}$ of end-of-cycle iterates; it lies in the model. The cycle map T (three successive rescalings) is continuous on K, and $L(T\boldsymbol{\mu}) \geq L(\boldsymbol{\mu})$ with equality only if every rescaling factor equals 1 (uniqueness of the block maximizer in step 2). The values $L(\mu^{(3s)})$ increase to a finite limit L^*; by continuity $L(\boldsymbol{\mu}^*) = L^*$ and $L(T\boldsymbol{\mu}^*) = \lim L(\mu^{(3s_r+3)}) = L^*$. Hence all factors at $\boldsymbol{\mu}^*$ equal 1: $\boldsymbol{\mu}^*$ matches the XY, XZ and YZ margins.

Step 5 (conclusion). By Birch's theorem there is only one such point, the MLE $\hat{\boldsymbol{\mu}}$. A bounded sequence all of whose limit points equal $\hat{\boldsymbol{\mu}}$ converges to $\hat{\boldsymbol{\mu}}$. The argument used only that each step rescales by a function of one generator's margin, so it applies to any hierarchical model. ■

Proposition 8.28 (IPF for direct models)

For independence in a two-way table, IPF started at $\mu^{(0)} \equiv 1$ reaches $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$ after one cycle.

Proof. $\mu^{(0)}_{i+} = J$, so $\mu^{(1)}_{ij} = 1 \cdot n_{i+}/J$. Then $\mu^{(1)}_{+j} = n/J$, so $\mu^{(2)}_{ij} = \frac{n_{i+}}{J} \cdot \frac{n_{+j}}{n/J} = \frac{n_{i+}n_{+j}}{n}$. This matches both margins, so further steps multiply by 1. ■

More generally IPF gives the ML fit of a direct model within one cycle (if the margins are fitted in a suitable order).

8.7.3 Comparison of iterative methods

IPF is simple, needs no matrix inversion, and converges even when the likelihood is poorly behaved (zero fitted counts, boundary estimates); it is feasible for very large tables. But it applies mainly to models whose likelihood equations match margins, may converge slowly, and does not yield parameter estimates or standard errors as by-products (these can be computed afterwards: parameters are contrasts of $\log \hat{\mu}$, and (8.15) gives the covariance). Newton–Raphson is general-purpose, fast near the optimum and yields $\widehat{\mathrm{cov}}(\hat{\boldsymbol{\beta}})$; it is what GLM software uses.

8.7.4 Contingency table standardization

Example 8.29 (Raking the abortion-attitude table)

A General Social Survey table of schooling by attitude toward legalized abortion was standardized (Smith, 1976) to have all row and column totals 100 while keeping all odds ratios. IPF: $\mu^{(0)}_{ij} = n_{ij}$ and, for $t = 1, 3, 5, \ldots$,

$$\mu^{(t)}_{ij} = \mu^{(t-1)}_{ij} \frac{100}{\mu^{(t-1)}_{i+}}, \qquad \mu^{(t+1)}_{ij} = \mu^{(t)}_{ij} \frac{100}{\mu^{(t)}_{+j}}.$$

Each step multiplies a whole row (column) by one constant, so odds ratios never change. Result (standardized values in parentheses):

Schooling	Disapprove	Middle	Approve
Less than high school	209 (49.4)	101 (32.0)	237 (18.6)
High school	151 (32.8)	126 (36.6)	426 (30.6)
More than high school	16 (17.8)	21 (31.3)	138 (50.9)

The standardized table shows a clear ridge on the main diagonal: more education goes with more favorable attitudes.

Standardization (*raking*) is used to compare tables with different margins and to adjust sample data to census margins. It corresponds to fitting $\log(E_{ij}/\mu_{ij}) = \lambda + \lambda^S_i + \lambda^A_j$, i.e. $\log E_{ij} - \log n_{ij} = \lambda + \lambda^S_i + \lambda^A_j$: the term $\log n_{ij}$ is an *offset*, a predictor with coefficient fixed at 1. In GLM software one enters pseudo-counts with the target margins and uses $\log n_{ij}$ as the offset (see Section 9.7).

8.8 R lab

Motivation. Loglinear models are Poisson GLMs on a data frame of cell counts, so `glm(..., family = poisson)` fits them; `loglin` and `MASS::loglm` fit them by IPF.

Enter the student survey with one row per cell; `A*C + A*M + C*M` is the symbol (AC, AM, CM).

```
acm <- expand.grid(M = c("yes","no"), C = c("yes","no"), A = c("yes","no"))
acm$count <- c(911, 538, 44, 456, 3, 43, 2, 279)
fit.hom <- glm(count ~ A*C + A*M + C*M, family = poisson, data = acm)
fit.ci  <- update(fit.hom, . ~ . - A:C)            # (AM, CM)
c(deviance(fit.hom), df.residual(fit.hom))         # G^2 = 0.37, df = 1
anova(fit.ci, fit.hom, test = "LRT")               # 187.4 on 1 df
exp(coef(fit.hom)["Ano:Cno"])                      # conditional AC OR = 7.8
exp(confint.default(fit.hom)["Ano:Cno", ])         # Wald CI (5.5, 11.0)
```

Fitted values and conditional odds ratios from the fitted table; compare a model's marginal AC table.

```
mu <- xtabs(fitted(fit.ci) ~ A + C + M, data = acm)
mu[1,1,1]*mu[2,2,1] / (mu[1,2,1]*mu[2,1,1])        # = 1 under (AM, CM)
mAC <- apply(mu, c(1,2), sum)
mAC[1,1]*mAC[2,2] / (mAC[1,2]*mAC[2,1])            # marginal AC OR = 2.7
```

IPF with `loglin` (margins listed by variable number) and `loglm`; both give $G^2 = 0.37$.

```
tab <- xtabs(count ~ A + C + M, data = acm)
ipf <- loglin(tab, margin = list(c(1,2), c(1,3), c(2,3)), fit = TRUE)
c(ipf$lrt, ipf$df)
MASS::loglm(~ A*C + A*M + C*M, data = tab)
```

Raking a table to margins 100 by direct IPF, and the four-way accident model (GLS, GI, IL, IS).

```
ab <- matrix(c(209,101,237, 151,126,426, 16,21,138), 3, byrow = TRUE)
m <- ab
for (it in 1:100) { m <- m * 100 / rowSums(m); m <- t(t(m) * 100 / colSums(m)) }
round(m, 1)                                        # 49.4 32.0 18.6 ...
maine <- expand.grid(I = c("no","yes"), S = c("no","yes"),
                     L = c("urban","rural"), G = c("female","male"))
maine$count <- c(7287,996, 11587,759, 3246,973, 6134,757,
                 10381,812, 10969,380, 6123,1084, 6693,513)
fit4 <- glm(count ~ G*L*S + G*I + L*I + S*I, family = poisson, data = maine)
c(deviance(fit4), df.residual(fit4))               # 7.5 on 4 df
```